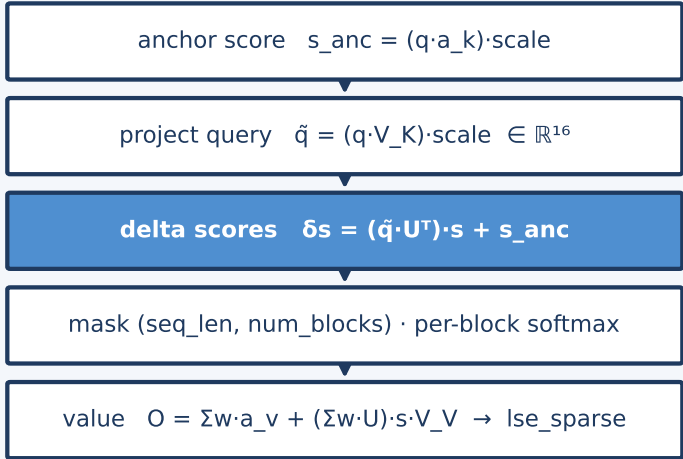


Fused Sparse Decode Attention (compute_decode_attention_static, mx.compile)

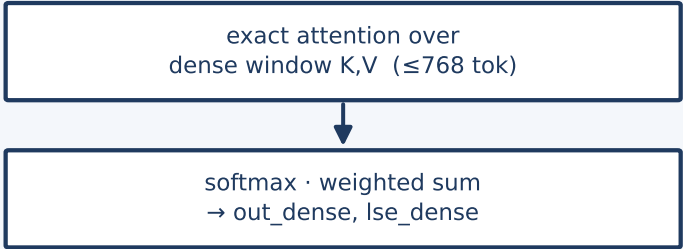
Query scored in low-rank space — KV is $[B, d]$; two branches merged by log-sum-exp

query q $[H, d]$

Sparse / compressed branch



Dense recency branch



$O(r \cdot B)$ per block — no $O(B \cdot d)$ decompression

flash-style LSE merge
 $w = \text{softmax}([lse_sparse, lse_dense])$
 $out = w_s \cdot O_sparse + w_d \cdot O_dense$

NaN/inf-guarded so a fully-masked branch contributes exactly zero weight